

Q.1 Drive an expression for time independent Schrodinger wave Equation? ①

Sol The Schrodinger wave Equation is the fundamental Equation of wave mechanics. In the same sense as the Newton's second law of motion of Classical mechanics.

It is the differential Equation of the de-broglie wave associated with particle and describe the motion of particle.

Time independent Schrodinger wave Equation:-

Let ψ be the wave function for de-broglie wave at location $\vec{r}$ at time t .

Then Equation of wave in 3-dimension can be written as

$$\nabla^2 \psi = \frac{1}{u^2} \frac{\partial^2 \psi}{\partial t^2} \quad \text{--- (1)}$$

where ∇ = operator $\left\{ \nabla = i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right\}$

u = wave velocity

The solution of Eq (1) we get

$$\psi = \psi_0 e^{-i\omega t} \quad \text{--- (2)}$$

where diff. of Eq (2) w.r. to time twice

$$\frac{\partial \psi}{\partial t} = \frac{\partial}{\partial t} (\psi_0 e^{-i\omega t})$$

ψ_0 = amplitude of wave
 ω = angular frequency
 $\omega = 2\pi\nu$

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t} (-i\omega)$$

$$\frac{\partial \psi}{\partial t} = -i\omega \psi_0 e^{-i\omega t}$$

again diff.

$$\frac{\partial^2 \psi}{\partial t^2} = -i\omega \psi_0 \frac{\partial}{\partial t} (e^{-i\omega t})$$

$$= -i\omega \psi_0 e^{-i\omega t} (-i\omega)$$

$$= i^2 \omega^2 \psi_0 e^{-i\omega t}$$

$$= -\omega^2 \psi_0 e^{-i\omega t}$$

$$\{ i^2 = -1 \}$$

$$\{ \psi = \psi_0 e^{-i\omega t} \}$$

$$\boxed{\frac{\partial^2 \psi}{\partial t^2} = -\omega^2 \psi} \quad \text{--- (3)}$$

Equating Eq (1) and (3) we get

$$\nabla^2 \psi = \frac{1}{u^2} [-\omega^2 \psi]$$

$$\nabla^2 \psi = -\frac{\omega^2}{u^2} \psi$$

$$\left\{ \begin{array}{l} \because \omega = 2\pi\nu \\ \omega^2 = 4\pi^2\nu^2 \end{array} \right\}$$

$$\nabla^2 \psi = -\frac{4\pi^2\nu^2}{u^2} \psi$$

$$\left\{ \begin{array}{l} \nu = u/\lambda \\ \nu^2 = u^2/\lambda^2 \end{array} \right\}$$

$$= -\frac{4\pi^2 u^2}{u^2 \times \lambda^2} \psi$$

$$\nabla^2 \psi = -\frac{4\pi^2 \psi}{\lambda^2} \quad \text{--- (4)}$$

by de-Broglie equation.

$$\lambda = \frac{h}{mv}$$

from eq (4)

$$\nabla^2 \psi = -\frac{4\pi^2}{\left(\frac{h}{mv}\right)^2} \psi \Rightarrow -\frac{4\pi^2 m^2 v^2}{h^2} \psi$$

$$\nabla^2 \psi = -\frac{4\pi^2 m^2 v^2}{4\pi^2 h^2} \psi$$

$$\left\{ \because h = 2\pi\hbar \right\}$$

$$\boxed{\nabla^2 \psi = -\frac{m^2 v^2}{\hbar^2} \psi} \quad \text{--- (5)}$$

if E & V be the total and potential energy of system then.

$$\text{total energy} = P.E + K.E$$

$$E = V + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = E - V$$

$$mv^2 = 2(E - V) \quad \left\{ \begin{array}{l} \text{multiply by } m \\ \text{both side} \end{array} \right\}$$

$$m^2 v^2 = 2m(E - V) \quad \text{--- (6)}$$

from (5) and (6)

$$\nabla^2 \psi = -\frac{2m(E - V)}{\hbar^2} \psi$$

$$\boxed{\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0}$$

this equation called Schrodinger time independent w. eq.

for free particle $V=0$

$$\text{So } \nabla^2 \psi + \frac{2m}{\hbar^2} (E-0) \psi = 0$$

$$\boxed{\nabla^2 \psi + \frac{2mE}{\hbar^2} \psi = 0}$$

This equation is called time independent Schrodinger wave eq. for free particle.

Q.2 Drive an expression for time dependent Schrodinger wave Equation?

Sol The Equation of wave in 3-D is written as

$$\nabla^2 \psi = \frac{1}{k^2} \frac{\partial^2 \psi}{\partial x^2} \quad \text{--- (1)}$$

The solution of this Equation is

$$\psi = \psi_0 e^{-i\omega t}$$

Differential of eq. w.r. to t

$$\frac{\partial \psi}{\partial t} = \psi_0 e^{-i\omega t} \times (-i\omega)$$

$$= -i\omega \psi_0 e^{-i\omega t}$$

$$\left\{ \psi_0 e^{-i\omega t} = \psi \right\}$$

$$\frac{\partial \psi}{\partial t} = -i\omega \psi$$

$$\left\{ \omega = 2\pi\nu \right\}$$

$$\frac{\partial \psi}{\partial t} = -2i\pi\nu \psi \quad \text{--- (2)}$$

we know the energy of photon

$$E = h\nu$$

$$\nu = \frac{E}{h}$$

from (2)

$$\frac{\partial \psi}{\partial t} = -2i\pi\nu\psi$$

$$\left\{ \nu = \frac{E}{h} \right\}$$

$$\frac{\partial \psi}{\partial t} = -\frac{2i\pi E\psi}{h}$$

$$\left\{ h = \frac{h}{2\pi} \right\}$$

$$= -\frac{i E \psi}{h}$$

$$\left\{ h = 2\pi\hbar \right\}$$

$$\frac{\partial \psi}{\partial t} = -\frac{i E \psi}{\hbar}$$

$$\frac{\partial \psi}{\partial t} = -\frac{i^2 E \psi}{i\hbar}$$

$$\left\{ i^2 = -1 \right\}$$

$$\frac{\partial \psi}{\partial t} = \frac{E \psi}{i\hbar} \Rightarrow i\hbar \frac{\partial \psi}{\partial t} = E \psi$$

$$\boxed{E\psi = i\hbar \frac{\partial \psi}{\partial t}} \quad \text{--- (3)}$$

time dependent Schrodinger wave equation
by time independent eq.

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$\nabla^2 \psi = -\frac{2m}{\hbar^2} (E - V) \psi$$

$$-\frac{\hbar^2}{2m} \nabla^2 \psi = (E - V) \psi$$

$$\boxed{V\psi - \frac{\hbar^2}{2m} \nabla^2 \psi = E\psi} \quad \text{--- (4)}$$

Equating equation (3) and (4)

$$E\psi = i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

This is also called time dependent Schrodinger wave equation.

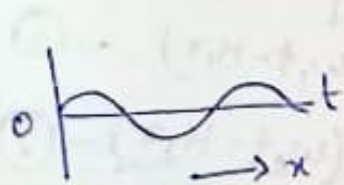
Q.4 Explain phase and group velocity? write relation between them? (3)

Ans Phase velocity / wave velocity :-

when a monochromatic wave (having single wavelength and single frequency) travels in a medium then the velocity of wave particle is called phase or wave velocity. it is denoted by V_p
OR

the velocity of single particle of single wave is called phase/wave velocity

Explanation:- Consider a wave travel in x direction then the displacement of wave is



$$y = a \sin(\omega t - kx)$$

where a = amplitude of wave

ω = angular frequency of wave

k = propagation constant of wave

the term $(\omega t - kx)$ is constant for plane

So

$$\omega t - kx = \text{constant}$$

difl.

$$\omega dt - k dx = 0$$

$$\omega dt = k dx$$

$$\frac{dx}{dt} = \frac{\omega}{k}$$

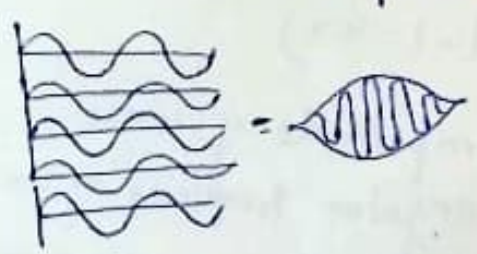
$$\left\{ \frac{dx}{dt} = \text{velocity} \right\}$$

phase velocity $\boxed{V_p = \frac{\omega}{k}}$

Group velocity :- group velocity is that velocity in which all waves are travel in group. it is denoted by v_g .
OR

The group velocity of wave is the velocity with which the overall shape of the wave amplitude (envelope) propagates through space.

Explanation :- Let us two waves having same amplitude a and different angular frequency ω_1 and ω_2 , propagation with propagation constant k_1 and k_2 then the displacement equation of wave.



$$y_1 = a \sin(\omega_1 t - k_1 x) \quad \text{--- (1)}$$

$$y_2 = a \sin(\omega_2 t - k_2 x) \quad \text{--- (2)}$$

By the principle of superposition wave

$$y = y_1 + y_2$$

$$y = a \sin(\omega_1 t - k_1 x) + a \sin(\omega_2 t - k_2 x)$$

$$y = a [\sin(\omega_1 t - k_1 x) + \sin(\omega_2 t - k_2 x)]$$

$$\left\{ \because \sin C + \sin D = 2 \sin\left(\frac{C+D}{2}\right) \cos\left(\frac{C-D}{2}\right) \right\}$$

$$y = 2a \left[\sin\left(\frac{(\omega_1 t - k_1 x) + (\omega_2 t - k_2 x)}{2}\right) \cos\left(\frac{(\omega_1 t - k_1 x) - (\omega_2 t - k_2 x)}{2}\right) \right]$$

$$(9) \quad y = 2a \sin \left[\frac{(\omega_1 + \omega_2)t - (k_1 + k_2)x}{2} \right] \cos \left[\frac{(\omega_1 - \omega_2)t - (k_1 - k_2)x}{2} \right]$$

$$\text{let } \frac{\omega_1 + \omega_2}{2} = \omega \quad \omega_1 - \omega_2 = \Delta\omega$$

$$\frac{k_1 + k_2}{2} = k \quad k_1 - k_2 = \Delta k$$

$$\text{so } y = 2a \sin(\omega t - kx) \cos \left(\frac{\Delta\omega t}{2} - \frac{\Delta k x}{2} \right)$$

the term $a \sin(\omega t - kx)$ represent the phase velocity
 & the amplitude of wave packet = $2a \cos \left(\frac{\Delta\omega t}{2} - \frac{\Delta k x}{2} \right)$

$$\text{so } \frac{\Delta\omega t}{2} - \frac{\Delta k x}{2} = \text{Constant}$$

$$\text{diff. } \frac{\Delta\omega dt}{2} - \frac{\Delta k dx}{2} = 0$$

$$\Delta\omega dt = \Delta k dx$$

$$\boxed{\frac{dx}{dt} = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk}}$$

Group velocity

$$\boxed{V_g = \frac{\Delta\omega}{\Delta k} = \frac{d\omega}{dk}}$$

Relation b/w phase/wave velocity and group velocity

Ans we know that

$$\text{phase velocity } V_p = \frac{\omega}{k}$$

$$\omega = k \cdot V_p$$

diff. of this eq. with respect to k

$$\frac{d\omega}{dk} = \frac{d}{dk} (k \cdot V_p)$$

$$\frac{dw}{dk} = k \cdot \frac{dv_p}{dk} + v_p \cdot 1$$

$$= v_p + k \cdot \frac{dv_p}{dk}$$

$$\left\{ \begin{aligned} k &= \frac{2\pi}{\lambda} \\ dk &= -\frac{2\pi}{\lambda^2} d\lambda \end{aligned} \right\}$$

$$\frac{dw}{dk} = v_p + \frac{2\pi}{\lambda} \cdot \frac{dv_p}{\left(-\frac{2\pi}{\lambda^2} d\lambda\right)}$$

$$= v_p - \frac{2\pi}{\lambda} \times \frac{\lambda^2}{2\pi} \frac{dv_p}{d\lambda}$$

$$\frac{dw}{dk} = v_p - \lambda \frac{dv_p}{d\lambda}$$

$$\boxed{v_g = v_p - \lambda \frac{dv_p}{d\lambda}}$$

This shows the relation between phase and group velocity.

different cases :-

case-1 if $\frac{dv_p}{d\lambda} = 0$ (if phase velocity does not depend on wavelength)

then

$$v_g = v_p - \lambda \frac{dv_p}{d\lambda}$$

$$v_g = v_p - \lambda(0)$$

$$\boxed{v_g = v_p}$$

The group velocity and the phase velocity are equal. Such medium is called non-dispersive medium. (1)

Case-2 in a dispersive medium
 $\frac{dv_p}{d\lambda}$ is positive
then $v_g < v_p$

for Relativistic particle

phase velocity $v_p = \frac{c^2}{v}$

group velocity $v_g = v$ $v_p v_g = c^2$

for non relativistic free particle

$$v_p = \frac{v_g}{2}$$

Q.5 state and proof Heisenberg uncertainty principle

Ans According to this principle

"It is impossible to measure/determine the exact position and momentum of a particle simultaneously"

proof:- In macroscopically it is possible to exactly measure the position of moving particle and the momentum of particle at any instant. but in microscopically it is not possible to exact measure the momentum and position of particle

simultaneously

(12)

The origin is that the behaviour of a moving particle is expressed by a wave packet.

according to max born

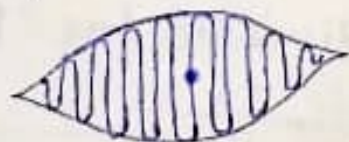
particle can be found any where within the wave packet, so that position of particle is uncertain within the limit of wave packet.

if wave packet is small the momentum of particle can^{not} be measure exactly but its position can be determined.



wave packet

if wave packet is large the position of particle can not be measure exactly but its momentum can be determined



wave packet

if Δx is the uncertainty in position & Δp is the uncertainty in momentum

then

the product of uncertainty in the measurement of momentum and position is of the order of $h/2\pi$

by this principle

$$\Delta x \cdot \Delta p \approx \frac{h}{2\pi}$$

$$\Delta x \cdot \Delta p = \frac{h}{2}$$

in exact form the product of uncertainties in position and momentum of particle can not less than $\frac{h}{2}$

$$\Delta x \cdot \Delta p \geq \frac{h}{2}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{2\pi\lambda}$$

$$\left\{ h = \frac{h}{2\pi} \right\}$$

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

this is called Heisenberg uncertainty principle.

Q.6 Explain dual nature of light?

The optical phenomenon, such as interference, diffraction, polarization of light could be explained by wave theory of light, but

wave theory of light failed to explain, photo-electric effect, Compton effect, absorption and emission of light radiation by substance.

These phenomena could be explained by quantum theory of light.

by Einstein, the energy of light is concentrated into small region. This represents the smallest quantity of energy called photon. Photon is an energy particle, hence light shows itself as wave nature at one hand and particle nature on other hand.

This shows that light is having wave nature as well as particle nature. This nature of light known as **dual nature of light**.

Q7 Derive an expression for wavelength of De-Broglie equation?

Ans According to de-broglie "a moving matter particle when a particle is moving, a wave is associated with that particle" a wave is called matter wave or De-Broglie wave.

the energy of photon of frequency ν and wave length λ is given by

$$E = h\nu \quad \text{--- (1)}$$

where $E =$ energy of photon
 $h =$ planck's constant
 $\nu =$ frequency of photon

if mass of particle is converted into energy the energy is given by Einstein mass-energy relation is given by

$$E = mc^2 \quad \text{--- (2)}$$

from (1) and (2)

$$h\nu = mc^2$$

$$\frac{hc}{\lambda} = mc^2$$

$$\frac{h}{\lambda} = mc$$

$$\lambda = \frac{h}{mc} \quad \text{--- (3)}$$

$$\left\{ \nu = \frac{c}{\lambda} \right\}$$

if photon travels speed of light then

$$c = v$$

(15)

from (3)

$$\lambda = \frac{h}{mv}$$

$$mv = p \text{ (momentum)}$$

$$\lambda = \frac{h}{p}$$

— (4)

This equation is called De-Broglie equation.
 λ is called De-Broglie wavelength

case-1

if the particle having mass m moving with velocity v , then ~~the~~

$$\text{kinetic energy} = \frac{1}{2}mv^2$$

$$= \frac{m \times mv^2}{2m} \Rightarrow \frac{m^2v^2}{2m} \quad \{mv = p\}$$

$$E = \frac{p^2}{2m}$$

$$2mE = p^2$$

$$p = \sqrt{2mE}$$

from Eq (4)

$$\lambda = \frac{h}{p} =$$

$$\lambda = \frac{h}{\sqrt{2mE}} \quad \text{--- (5)}$$

Case-II

According to the kinetic theory of gases, then the average kinetic energy of the material particle is

$$E = \frac{1}{2}mv^2 = \frac{3}{2}KT$$

$$E = \frac{3}{2}KT$$

K = Boltzmann's constant = $1.38 \times 10^{-23} \text{ J/K}$

T = absolute temperature

from eq (5)

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2m \times \frac{3KT}{2}}}$$

$$\lambda = \frac{h}{\sqrt{3mKT}}$$

$$\boxed{\lambda = \frac{h}{\sqrt{3mKT}}} \quad \text{--- (6)}$$

Case-III

Suppose an electron accelerates through a potential difference V (volt)

then work done by electric = gain in kinetic energy

$$eV = \frac{1}{2}mv^2$$

$$2eV = mv^2$$

$$2eV \times m = m \times (mv^2)$$

$$2meV = m^2v^2$$

$$mv = \sqrt{2meV}$$

by De-Broglie wavelength of e^-

(19)

$$\lambda = \frac{h}{mv}$$

$$\lambda = \frac{h}{\sqrt{2meV}}$$

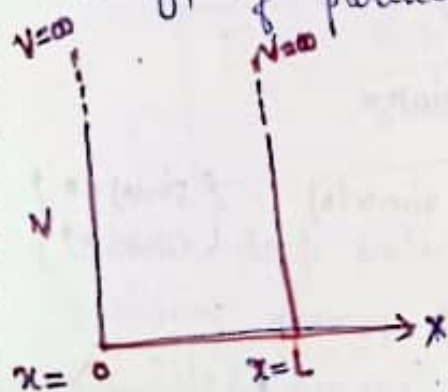
Particle in one dimensional Box:-

consider a particle moving inside a box in x -direction. the particle is bouncing back and forth between the wall of the box. let L be the width of box. & V be the potential energy of particle. V is infinite on both sides of box.

$$V = 0 \text{ for } 0 < x < L$$

$$V = \infty \text{ for } x \leq 0 \text{ and } x \geq L$$

The particle cannot exist outside the box.



by time independent schrodinger wave eq.

in one dimension

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - V_0) \psi = 0 \quad \text{--- (1)}$$

$$V=0 \quad \frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} (E - 0) \psi = 0$$

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{2m}{\hbar^2} E \psi = 0 \quad \text{--- (2)}$$

$$\text{Let } k^2 = \frac{2mE}{\hbar^2}$$

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

from eq (2) we get

$$\frac{\partial^2 \psi}{\partial x^2} + k^2 \psi = 0 \quad \text{--- (3)}$$

the solution of eq. --- (3)

$$\psi = A \cos kx + B \sin kx \quad \text{--- (4)}$$

at boundary condition.

particle can not exist outside the box, so its wave function ψ is 0 for $x=0$

$$x=0, \psi=0$$

from eq (4)

$$\psi = A \cos kx + B \sin kx$$

$$0 = A \cos k(0) + B \sin k(0) \quad \left\{ \begin{array}{l} \sin(0) = 0 \\ \cos(0) = 1 \end{array} \right\}$$

$$0 = A \times 1 + B \times 0$$

$$0 = A + 0$$

$$\boxed{A = 0}$$

from (4)

$$\psi = 0 \cdot \cos kx + B \sin kx$$

$$\boxed{\psi = B \sin kx} \quad \text{--- (5)}$$

at boundary condition.

$$x=L, \psi=0$$

from (5)

$$\psi = B \sin kL$$

$$0 = B \sin kL$$

$$\boxed{B \neq 0}$$

$$\sin KL = 0$$

(19)

$$\left\{ \sin n\pi = 0 \right\}$$

$$\sin KL = \sin n\pi$$

$$KL = n\pi$$

$$K = \frac{n\pi}{L}$$

$$K^2 = \frac{n^2\pi^2}{L^2}$$

$$\left\{ \because K^2 = \frac{2mE}{\hbar^2} \right\}$$

$$\frac{2mE}{\hbar^2} = \frac{n^2\pi^2}{L^2}$$

$$E = \frac{\hbar^2 n^2 \pi^2}{2mL^2} = \frac{\hbar^2 n^2 \pi^2}{4\pi^2 \times 2mL^2}$$

$$\left\{ \hbar = \frac{h}{2\pi} \right\}$$

$$E_n = \frac{n^2 h^2}{8mL^2}$$

eigen energy.

in ∞ deep potential well particle can not have an arbitrary energy, but can have discrete energy. each permitted energy is called eigen value of particle.

eigen wave function

$$\int_0^L \psi \psi^* dx = 1$$

$$\left\{ \psi = B \sin \frac{n\pi x}{L} \right\}$$

from (5)

$$\int_0^L B \sin \frac{n\pi x}{L} \cdot B^* \sin \frac{n\pi x}{L} dx = 1$$

$$B^2 \int_0^L \sin^2 \left(\frac{n\pi x}{L} \right) dx = 1$$

$$\left\{ \sin^2 \theta = 1 - \frac{\cos 2\theta}{2} \right\}$$

$$B^2 \int_0^L \left(1 - \frac{\cos 2 \left(\frac{n\pi x}{L} \right)}{2} \right) dx = 1$$

$$\frac{B^2}{2} \left[\int_0^L dx - \int_0^L \cos 2 \left(\frac{n\pi x}{L} \right) dx \right] = 1$$

$$\frac{B^2}{2} \left\{ \left[x \right]_0^L - \left[\frac{\sin 2 \left(\frac{n\pi x}{L} \right)}{\frac{n\pi}{L}} \right]_0^L \right\} = 1$$

$$\frac{B^2}{2} \left\{ [L-0] - 0 \right\} = 1$$

$$\frac{B^2 L}{2} = 1$$

$$B^2 = \frac{2}{L} \Rightarrow B = \sqrt{\frac{2}{L}}$$

from (5)

$$\psi = B \sin \frac{n\pi x}{L}$$

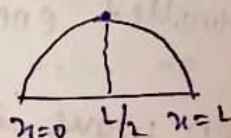
$$\boxed{\psi = \sqrt{\frac{2}{L}} \sin \frac{n\pi x}{L}}$$

Graph. for first excited state $n=1$,

$$\sin \frac{n\pi x}{L} = \sin \frac{\pi}{2}$$

$$\frac{n\pi x}{L} = \frac{\pi}{2}$$

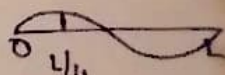
$$x = \frac{L}{2}$$



for 2nd excited state $n=2$

$$\sin \frac{2\pi x}{L} = \sin \frac{\pi}{2}$$

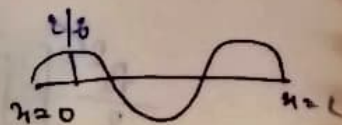
$$\frac{2\pi x}{L} = \frac{\pi}{2} \Rightarrow x = \frac{L}{4}$$



for 3rd excited state $n=3$

$$\sin \frac{3\pi x}{L} = \sin \frac{\pi}{2}$$

$$\boxed{x = \frac{L}{6}}$$



wave function:- A wave function is a mathematical function that relates the location of an electron. (position, spin, momentum, and time). (21)

wave function can help, the probability of finding an e^- within the matter wave can be explained. it is denoted by ψ .

properties of wave function:- ψ must be continuous, and single valued everywhere.

- * ψ must be normalized, which means that ψ must go to 0 as $x \rightarrow \pm \infty$, $y \rightarrow \pm \infty$, $z \rightarrow \pm \infty$
- * the probability of finding particle if it exist is 1.
- * all measurable information about the particle is available.
- * By using schrodinger wave equation energy calculation become easy.

operator:- operators is a mathematical entities which are used to represent physical process that result in the change of the state vector of the system.

- ① energy operator
- ② momentum operator

① Total energy operator :- By using time independent schrodinger wave equation. (28)

$$\nabla^2 \psi + \frac{2m}{\hbar^2} (E - V) \psi = 0$$

$$\nabla^2 \psi = - \frac{2m}{\hbar^2} (E - V) \psi$$

$$- \frac{\hbar^2}{2m} \nabla^2 \psi = E\psi - V\psi$$

$$V\psi - \frac{\hbar^2}{2m} \nabla^2 \psi = E\psi$$

$$E\psi = \left(-\frac{\hbar^2}{2m} \nabla^2 + V \right) \psi$$

$E = -\frac{\hbar^2}{2m} \nabla^2 + V$
$E = i\hbar \frac{\partial}{\partial t}$

Energy operator called Hamiltonian operator.

② momentum operator :- By using time independent schrodinger wave eq.

$$E = i\hbar \frac{\partial}{\partial t} \quad E = -\frac{\hbar^2}{2m} \nabla^2 + V$$

$$K.E + V = -\frac{\hbar^2}{2m} \nabla^2 + V$$

$$\{ E = K.E + V \}$$

$$\{ K.E = \frac{1}{2} m v^2 \}$$

$$\frac{1}{2} m v^2 = -\frac{\hbar^2}{2m} \nabla^2$$

$$m^2 v^2 = -\hbar^2 \nabla^2$$

$$(m v)^2 = \frac{\hbar^2}{i^2} \nabla^2$$

$$\left\{ \begin{array}{l} m v = p \\ -1 = i^2 \end{array} \right\}$$

$$p^2 = \left(\frac{\hbar}{i} \nabla \right)^2$$

(23)

$$p = \frac{\hbar}{i} \nabla$$

in components

$$p_x = \frac{\hbar}{i} \frac{\partial}{\partial x}$$

$$p_y = \frac{\hbar}{i} \frac{\partial}{\partial y}$$

$$p_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

} momentum operator

③ Kinetic Energy operator:-

We know that-

Total Energy = Kinetic Energy + Potential Energy

$$E = K.E + V$$

$$\left\{ E = -\frac{\hbar^2}{2m} \nabla^2 + V \right\}$$

$$-\frac{\hbar^2}{2m} \nabla^2 + V = K.E + V$$

$$K.E = -\frac{\hbar^2}{2m} \nabla^2$$